1. Telemetry Parameters (Input Features)

Power System

Parameter	Description
Battery Voltage	Spacecraft battery voltage
Battery Current	Charge/Discharge current
Battery SOC	State of Charge (%)
Battery Temperature	Battery health
Solar Panel Voltage	Solar array output
Solar Panel Current	Solar generation
Power Consumption	Total load
Power Generation	Total generated power

Thermal System

Parameter	Description
Payload Temperature	Payload health
CPU Temperature	Computer temperature
Battery Temperature	Battery heating
Solar Panel Temperature	Panel health
Spacecraft Internal Temperature	Cabin/System temperature
External Temperature	Space environment

Communication System

Parameter	Description
Signal Strength	
Downlink Rate	
Uplink Rate	
Packet Loss	
Latency	
Communication Window Status	

Attitude & Navigation (ADCS)

Parameter	Description
Roll	
Pitch	
Yaw	
Angular Velocity	
Reaction Wheel Speed	
Gyroscope X	
Gyroscope Y	
Gyroscope Z	
Magnetometer	
Star Tracker Status	

Orbit Parameters

Parameter	Description
Altitude	
Velocity	
Latitude	
Longitude	
Orbital Phase	
Eclipse Status	

Propulsion

Parameter	Description
Fuel Remaining	
Thruster Temperature	
Thruster Status	
Fuel Pressure	
Burn Duration	

Payload Parameters

Parameter	Description
Camera Status	
Instrument Temperature	
Instrument Power	
Data Collection Rate	
Payload Mode	

Computer Health

Parameter	Description
CPU Usage	
RAM Usage	
Storage Usage	
Process Health	
Software Version	

---

2. Mission Parameters

These are needed by the planner.

Parameter
Mission Goal
Mission Phase
Priority
Remaining Time
Observation Window
Communication Window
Resource Availability
Safe Mode Status

---

3. Constraint Parameters

The planner must verify these.

Battery > 30%

Temperature < 70°C

Fuel > 10%

CPU < 90%

Storage < 95%

Communication Available

Thruster not active during imaging

Power Budget maintained

---

4. Fault Injection Parameters

These are used during testing.

Battery Failure

Solar Panel Failure

Camera Failure

Sensor Drift

GPS Failure

Communication Delay

Packet Loss

Thermal Spike

CPU Overload

Memory Leak

Reaction Wheel Failure

Thruster Failure

Power Loss

Conflicting Sensors

Telemetry Missing

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5. ML Features (Feature Vector)

Example

Battery_Voltage

Battery_Current

Battery_SOC

Solar_Voltage

Solar_Current

Power_Load

Power_Generation

CPU_Temp

Payload_Temp

Battery_Temp

Altitude

Velocity

Fuel_Level

Thruster_Temp

Signal_Strength

Packet_Loss

CPU_Usage

RAM_Usage

Storage_Usage

Roll

Pitch

Yaw

Reaction_Wheel_Speed

Mission_Phase

Communication_Window

Observation_Window

These become the **X (features)**.

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6. ML Output Labels

Classification

Healthy

Battery Failure

Solar Panel Failure

Communication Failure

Thermal Anomaly

Power Anomaly

Sensor Failure

Propulsion Failure

Attitude Control Failure

Critical Failure

Safe Mode Required

Regression

Predict

Remaining Battery Life

Temperature after 30 min

Fuel Remaining

Mission Completion Probability

Time to Failure

Resource Utilization

Power Consumption

Communication Delay

---

7. Explainable AI Output

Instead of only saying

Battery Failure

Output

Failure:

Battery Failure

Confidence:

96%

Evidence:

Battery Voltage ↓

Battery Current ↑

Temperature ↑

Solar Generation ↓

Recommended Procedure:

Enter Safe Mode

Estimated Risk:

Medium

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8. Recommended ML Models

Task	**Model**
Anomaly Detection	Isolation Forest ⭐
Unknown Anomalies	Autoencoder ⭐⭐⭐
Failure Classification	XGBoost
Time-Series Forecasting	LSTM / Temporal Convolutional Network (TCN)
Resource Prediction	XGBoost Regressor
Root Cause Analysis	Bayesian Network + LLM
Explainability	SHAP / LIME
Planning	Google OR-Tools (not ML)

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9. Suggested Dataset Columns

Timestamp

Mission_Phase

Battery_Voltage

Battery_Current

Battery_SOC

Solar_Voltage

Solar_Current

Power_Load

Power_Generation

Battery_Temperature

CPU_Temperature

Payload_Temperature

Fuel_Level

Thruster_Status

Thruster_Temperature

Altitude

Velocity

Roll

Pitch

Yaw

Signal_Strength

Packet_Loss

Latency

CPU_Usage

RAM_Usage

Storage_Usage

Camera_Status

Communication_Window

Observation_Window

Fault_Type

Anomaly_Score

Failure_Class

Confidence

Recommended_Action

Final ML Pipeline

Telemetry Generator

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Data Preprocessing

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Feature Engineering

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▼

Isolation Forest

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▼

Anomaly Detection

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▼

XGBoost Classifier

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Failure Classification

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▼

LSTM / TCN

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Failure Prediction

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▼

LLM + RAG

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Root Cause Analysis

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Procedure Recommendation

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▼

Safety Verification

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▼

Human Approval

This parameter set is comprehensive enough for a **mission-grade telemetry dataset** and directly supports the planner, anomaly detector, predictive models, explainable AI, and operator decision support envisioned by the problem statement.